Ordering Is Not Invariant: The Absence of Permutation Group Structure in Language Model Representations

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Abstract

We investigate whether transformer language models develop internal representations with permutation group symmetry. Using operator fitting, group relation testing, and representation-theoretic analysis across two scales of Qwen2.5 (0.5B, 1.5B), we find that models are strongly permutation-*sensitive*—linear operators mapping between permuted representations fit 21× better than chance—but exhibit zero group-algebraic structure. Composition errors for S_3 operators are indistinguishable from random baselines. This null result holds across all layers, in PCA subspaces, in sparse autoencoder feature bases, and at both model scales. A positive control on synthetic data with exact group structure confirms the pipeline would detect it if present. We extend these findings to fact-level permutations, showing that reordering the same facts changes model representations 35–60% as much as presenting entirely different facts, with next-token predictions changing 30% of the time. We argue this reflects a fundamental property: local optimization under next-token prediction learns *functional* equivariance (correct outputs under transformation) without *structural* equivariance (group representations in latent space). This has implications for synthetic data generation, where the implicit assumption that semantic content is order-invariant may destroy information the model has learned about salience and discourse structure.

1 Introduction

Symmetry has been among the most powerful organizing principles in science. In physics, the discovery that physical laws are invariant under specific group actions—translations, rotations, gauge transformations—has led to conservation laws, unified theories, and deep structural understanding [1, 2]. In machine learning, the geometric deep learning program has shown that building symmetry into neural network architectures through equivariant layers improves sample efficiency and generalization [3, 4].

A natural question arises: do neural networks trained on data with implicit symmetries develop *internal* representations that respect those symmetries, even without explicit architectural constraints? Recent work on Group Crosscoders [5] provides a positive answer for vision: a convolutional network trained on natural images develops approximate dihedral group (D_{32}) equivariance in its early vision layers, with curve detectors exhibiting full rotational equivariance and line detectors showing 180° invariance.

We ask whether analogous structure exists in language. Specifically: do language model representations of permuted entities or reordered facts carry the algebraic structure of the symmetric group S_n? If "Alice gave Bob the book" and "Bob gave Alice the book" produce activations related by a linear operator W_π, do these operators compose as group elements—does $W_{\pi_1} W_{\pi_2} \approx W_{\pi_1 \circ \pi_2}$?

This question has practical significance beyond mechanistic interpretability. Modern approaches to synthetic data generation, such as the Simula framework [6], decompose data domains into independent factors sampled from hierarchical taxonomies. This product-space decomposition implicitly assumes that reordering or recombining content is semantically neutral—that the "meaning" of a fact set is invariant under permutation of its elements. If language models have learned that ordering encodes information—salience, narrative priority, discourse structure—then synthetic data approaches that scramble ordering may destroy signal the model has learned to use.

Our contributions are:

1. A comprehensive empirical demonstration that language model representations are permutation-*sensitive* but not permutation-*equivariant*: linear operators fit well individually but fail all group composition tests (S_3 relation errors ≈ 1.0, indistinguishable from random baselines).
2. Positive controls and random baselines establishing that the detection pipeline works (relation errors $\to 0$ on synthetic group-structured data) and that the null result is not an artifact of insufficient signal.
3. Evidence that fact-level reordering is *not* a symmetry of the model's world representation: reordering the same facts changes activations 35–60% as much as presenting entirely different facts, with next-token predictions changing 30% of the time.
4. A conceptual framework distinguishing *functional equivariance* (correct outputs under transformation) from *structural equivariance* (group representations in latent space), explaining why the former is achievable without the latter under local optimization.

2 Background

2.1 Group Representations and Equivariance

A *representation* of a group G on a vector space V is a homomorphism $\rho : G \to GL(V)$—a map assigning to each group element g an invertible linear transformation $\rho(g)$ such that $\rho(g_1 g_2) = \rho(g_1)\rho(g_2)$. A function $f : V \to W$ is *equivariant* with respect to representations ρ_V, ρ_W if $f(\rho_V(g) \cdot x) = \rho_W(g) \cdot f(x)$ for all $g \in G, x \in V$.

The symmetric group S_n consists of all permutations of n elements. S_3 has order 6 and is generated by a transposition $s = (0\ 1)$ and a 3-cycle $r = (0\ 1\ 2)$ with presentation $\langle s, r \mid s^2 = e, r^3 = e, srs = r^{-1} \rangle$. Its irreducible representations are: the trivial representation (1D, all elements $\to 1$), the sign representation (1D, even permutations $\to +1$, odd $\to -1$), and the standard representation (2D).

We distinguish two notions of equivariance in neural networks:

- **Functional equivariance**: the model's *output* transforms correctly under input permutations. If we swap Alice and Bob in a prompt, the model's completion swaps accordingly.
- **Structural equivariance**: the model's *internal representations* carry group structure. The operators relating permuted activations satisfy group composition laws.

A model can achieve functional equivariance without structural equivariance—by implementing a lookup table rather than an algebra.

2.2 Prior Work

Group structure in vision models. Bhaskar et al. [5] introduced Group Crosscoders, training sparse autoencoders on concatenated activations from transformed inputs to discover symmetry structure. Applied to InceptionV1, they found approximate D_{32} equivariance in early vision features. This positive result establishes that group structure *can* emerge in neural networks trained on data with physical symmetries.

Geometric deep learning. Bronstein et al. [3] survey the program of building symmetry into network architectures. Cohen and Welling [4] showed that group equivariant CNNs improve generalization. Xu et al. [8] proved that vanilla transformers possess inter- and intra-token permutation equivariance—an architectural property distinct from learned representational structure.

Linear representations in LLMs. Park et al. [7] formalized the linear representation hypothesis using counterfactual semantics, showing that high-level concepts are represented as linear directions in LLM representation spaces. Our operator fitting approach is consistent with this framework: we test whether permutation-induced changes are linear (they are) and whether the resulting linear operators have algebraic structure (they do not).

Synthetic data generation. Davidson et al. [6] introduced Simula, decomposing data generation into coverage, complexity, and quality axes via hierarchical taxonomies sampled as a product space $\mathcal{T}_0 \times \mathcal{T}_1 \times \cdots \times \mathcal{T}_K$. This assumes factors are independently combinable—a form of exchangeability assumption that our results call into question.

Information structure in linguistics. The pragmatics literature has long established that utterance ordering encodes topic-focus structure, givenness, and information status [11, 12]. Given information typically precedes new information; propositions closest to the discourse topic are uttered first [13]. Our empirical findings are consistent with these linguistic principles.

3 Methods

3.1 Models and Setup

We study Qwen2.5-0.5B (24 layers, $d_{\text{model}} = 896$) and Qwen2.5-1.5B (28 layers, $d_{\text{model}} = 1536$), both base (non-instruct) models [10]. Activations are captured using TransformerLens [9] at the residual stream (post-block) of every layer, at the last token position.

3.2 Permutation Prompt Design

We construct two types of permutation experiments:

Entity-level permutations. Fixed sentence templates with named entity slots, e.g., "{e0} and {e1} entered the room. The first to enter was". Entity names (Alice, Bob, Carol) are chosen for single-token encoding and matched tokenization length. For S_3, we generate all 6 permutations of 3-entity templates across 10 templates and 10 entity sets, yielding 100 samples per non-identity permutation.

Fact-level permutations. Multi-sentence sequences where each sentence is an independent fact. We generate all $n!$ orderings and compare activations and predictions. We use 15 fact sets each for S_2 (2-fact) and S_3 (3-fact) sequences, including examples spanning urgency levels (e.g., "The room is on fire." alongside "It is Wednesday.").

3.3 Operator Fitting

For each permutation $\pi \in S_n$ (excluding identity), we collect paired activations $(h(x), h(\pi \cdot x))$ across many prompts and fit a linear operator W_π via least-squares regression:

$$W_\pi = \arg\min_W \|W \cdot H_{\text{base}}^T - H_{\text{perm}}^T\|_F^2 \tag{1}$$

where $H_{\text{base}}, H_{\text{perm}} \in \mathbb{R}^{n \times d}$ are matrices of base and permuted activations. We use an 80/20 train/test split and report test-set relative error $\|Y - \hat{Y}\|_F / \|Y\|_F$ and cosine similarity.

3.4 Group Relation Testing

Given fitted operators $\{W_\pi\}$, we test the S_3 presentation relations:

$$W_s^2 \approx I \quad \text{(transposition is involution)} \tag{2}$$

$$W_r^3 \approx I \quad \text{(3-cycle cubes to identity)} \tag{3}$$

$$W_s W_r W_s \approx W_r^{-1} = W_r^2 \quad \text{(conjugation relation)} \tag{4}$$

and full composition closure: $W_{\pi_1} W_{\pi_2} \approx W_{\pi_1 \circ \pi_2}$ for all pairs. We report relative error $\|LHS - RHS\|_F / \|RHS\|_F$.

3.5 Controls

Positive control. We construct synthetic activations with exact S_3 group structure: random base vectors $x \in \mathbb{R}^d$ mapped by exact representation matrices W_π (the combined trivial + sign + standard irrep, embedded in $d = 10$ dimensions via random orthogonal rotation) plus calibrated Gaussian noise. This validates that our pipeline detects group structure when present.

Random baseline. We fit operators on *shuffled* prompt pairs—base activations from one prompt paired with "permuted" activations from an unrelated prompt. This establishes the chance-level floor for both fit quality and group relation errors.

3.6 Additional Analyses

Subspace search. PCA on activation differences $\Delta h = h(\pi \cdot x) - h(x)$ to find a permutation-sensitive subspace. Operators fitted and group relations tested in projected subspaces of dimension 3–200.

Irreducible representation search. Direct optimization for the sign representation (direction v with $W_s v \approx -v$, $W_r v \approx +v$) and the standard 2D irrep.

SAE feature-space analysis. Sparse autoencoders (8× expansion, 5M training tokens) trained on layers 0 and 21. Activations encoded through the SAE, operators fitted and group relations tested in the sparse feature basis.

Fact-order analysis. For fact-level permutations: (1) relative activation distance between reorderings vs. between different fact sets; (2) KL divergence and top-1 agreement of next-token predictions; (3) group structure testing on fact-reordering operators.

4 Results

4.1 Entity-Level Permutations: Strong Signal, No Structure

Behavioral validation. The model reliably tracks entity permutations at the output level: swapping Alice and Bob in "{e0} and {e1} entered the room. The first to enter was" shifts the top-1 prediction from "Alice" to "Bob" with comparable confidence. KL divergences between identity and permuted outputs range from 0.15 to 2.1 across templates, confirming functional equivariance.

Activation alignment. Operator fit quality follows a U-shaped layer profile (Figure 1a, Table 1). Early layers (0–2) fit best, with test-set relative errors as low as 0.08 for the 0.5B model. Middle layers (10–16) fit worst (∼0.43). Late layers (20–22) show a secondary peak (∼0.26). The 1.5B model exhibits the same pattern (best at layer 0, test error 0.12). The permutation signal is strongly linearly predictable at all layers.

Group relations: the null result. Despite good individual operator fits, all S_3 presentation relations fail (Table 1). Relation errors are ≈ 1.0 at every layer, for both model scales. The positive control demonstrates that the pipeline detects group structure when present: at noise levels yielding comparable test error (∼0.1), relation errors are 0.04–0.06—twenty times lower than the real model's ∼1.0. The random baseline confirms that the permutation signal is real (operators fit 21× better than shuffled pairs) but the algebraic structure is not (relation errors are comparable to random).

The pattern is robust. We tested every natural place group structure might hide:

- **PCA subspace** (3–200D): Best result at layer 0, dim 3: $s^2 = e$ error 0.09, but $r^3 = e$ still fails at 0.73.
- **Irrep search**: No sign representation or standard 2D irrep detected at any layer.

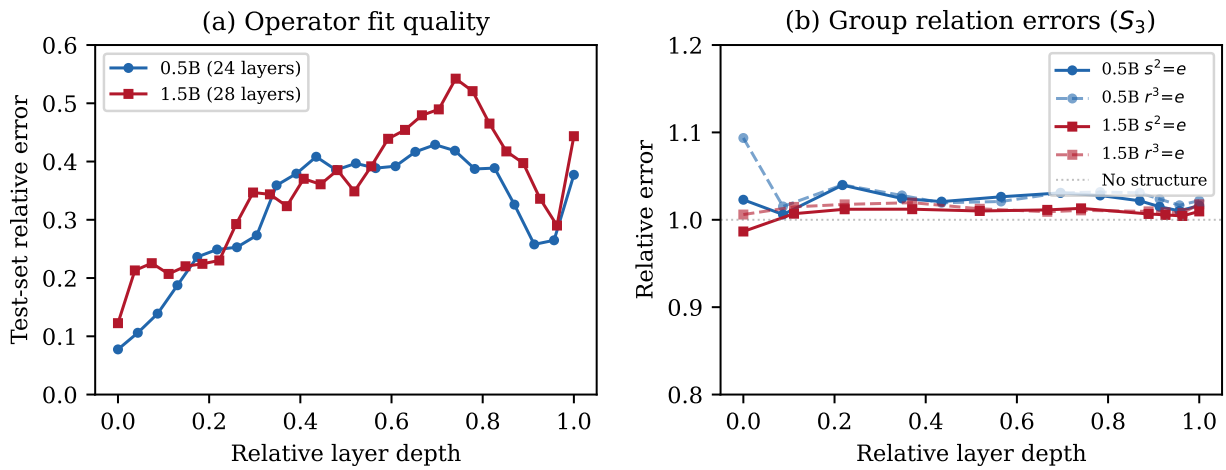

Figure 1: Layer-by-layer analysis for S_3 entity permutations. (a) Operator fit quality (test-set relative error) follows a U-shaped profile: best at early and late layers. Both model scales show the same pattern. (b) Group relation errors remain ≈ 1.0 at every layer regardless of fit quality, indicating no algebraic structure. The "no structure" baseline (gray dotted) at 1.0 represents the error expected from unrelated operators.

- **SAE feature space**: Group relations *worse* than raw activation space (errors 1–34×).
- **Eigenvalue analysis**: Transposition operators have eigenvalues clustered near $+1$ with none near -1 and det ≈ 0—they are rank-deficient projections, not reflections.
- **Scale**: The 1.5B model shows identical patterns across all 28 layers.

4.2 Controls

Positive control. Synthetic activations with exact S_3 representations embedded in $\mathbb{R}^{10}$ yield relation errors scaling linearly with noise (Figure 2, Table 2). At $\sigma = 0$ errors are numerically zero; at $\sigma = 0.1$ (comparable real-model test error) they are $\sim$0.05. The real model's errors ($\sim$1.0) correspond to the regime where group structure is completely destroyed ($\sigma \geq 1.0$). This confirms the pipeline works and that the null result is genuine.

Random baseline. At layer 0, shuffled-pair operators fit with test error 1.68 (vs. 0.08 for real pairs—a 21× ratio). But group relation errors from shuffled pairs are comparable to real pairs: $s^2 = e$ error 36.9 (shuffled) vs. 1.02 (real). The real model's relation errors, while lower than random, are still $\sim$1.0—firmly in the "no structure" regime. The signal that makes real operators fit well does not carry algebraic coherence.

4.3 Fact-Level Permutations: Ordering Encodes Information

Activation distance. Reordering the same facts changes last-token activations substantially (Figure 3, Table 3). The within-set relative distance (same facts, different order) averages 35–60% of the between-set distance (entirely different facts). This ratio is stable across layers and consistent for 2-fact and 3-fact sequences. The model does *not* treat fact-sets as permutation-invariant.

Behavioral impact. Next-token predictions change meaningfully with fact order: top-1 agreement is only 67% for 2-fact and 71% for 3-fact sequences (Table 4). Mean KL divergence is 0.35 (S_2) and 0.28 (S_3). The model produces *different continuations* for the same facts in different orders—confirming that ordering carries semantic weight.

Table 1: Operator fit quality and group relation errors for S_3 across selected layers. Test error measures how well W_π predicts held-out permuted activations. Group relation errors measure deviation from the S_3 presentation. An error of ≈ 1.0 indicates no better than random composition.

	Qwen2.5-0.5B			Qwen2.5-1.5B		
Layer	Test Err	$s^2 = e$	$r^3 = e$	Test Err	$s^2 = e$	$r^3 = e$
0	0.078	1.023	1.094	0.123	0.987	1.006
5	0.249	1.040	1.040	0.224	1.012	1.018
10	0.408	1.021	1.020	0.323	1.012	1.020
16	0.429	1.031	1.031	0.439	1.011	1.009
21	0.258	1.015	1.024	0.521	1.013	1.011
Positive control ($\sigma = 0.1$, $d = 10$)						
—	0.101	0.045	0.061			
Random baseline (shuffled pairs, layer 0)						
—	1.675	36.9	158.1			

Table 2: Positive control: S_3 group relation errors in synthetic data with exact group structure at varying noise levels ($d = 10$, $n = 200$ samples). Compare to real model errors of ≈ 1.0.

Noise σ	Test Err	$s^2 = e$	$r^3 = e$	$srs = r^{-1}$	Composition
0.00	0.000	0.000	0.000	0.000	0.000
0.01	0.010	0.005	0.006	0.006	0.004
0.10	0.101	0.045	0.061	0.060	0.044
0.50	0.465	0.217	0.329	0.290	0.217
1.00	0.752	0.425	0.679	0.631	0.435
Real model (layer 0): 0.078 test err, but $\sim$1.0 relation errors					

Case study. Consider three facts of varying urgency: (F1) "The room is on fire." (F2) "I am sitting at my desk researching internal representations in LLMs." (F3) "It is Wednesday." The KL divergence matrix across all 6 orderings (Figure 4) reveals: orderings ending with the same fact produce similar predictions (KL = 0.026 between orderings F-W-R and W-F-R, both ending with the research fact), while orderings ending with different facts diverge substantially (KL = 0.35 between F-R-W and R-W-F). The model attends most to the *last-mentioned* fact—a recency effect, not a symmetry.

Group structure. Fact-reordering operators show the same null result as entity permutations: good individual fits (test error 0.13 at layer 0) but $s^2 = e \approx 0.99$, $r^3 = e \approx 1.00$, composition errors ≈ 0.7–0.8. The pattern generalizes: *no* permutation operation on text—whether entity swaps or fact reordering—induces group-structured representations.

5 Discussion

5.1 Why No Group Structure?

Three candidate explanations compete:

(A) The territory. Idea space does not have S_n symmetry. Ordering genuinely encodes information—salience, temporal priority, narrative frame. The model correctly learned the world's asymmetry. This is supported by our fact-reordering results and by the pragmatics literature on information structure [11, 12].

(B) The map. Transformers cannot learn group structure even if it exists. But Group Crosscoders [5] refute the strong version: vision models *do* learn approximate group structure when the physical world has

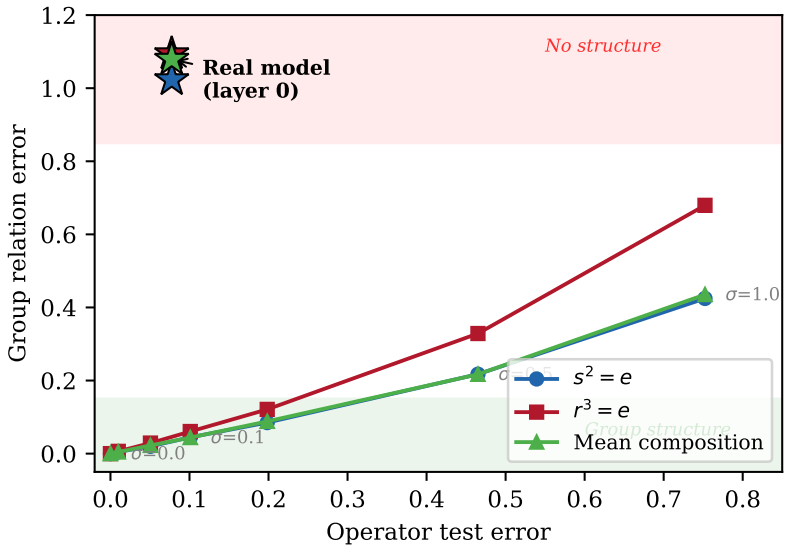

Figure 2: Positive control: group relation errors vs. operator test error for synthetic data with exact S_3 structure at varying noise levels. Lines show the calibration curve; stars mark the real model (layer 0). The real model sits in the "no structure" region (top) despite having low test error (left)—a pattern impossible under true group structure.

Table 3: Impact of fact reordering on model representations. "Within" = same facts in different order. "Between" = entirely different facts. Ratio $\ll 1$ would indicate permutation invariance; observed ratios of 0.35–0.60 show ordering substantially affects representations.

	2-fact (S_2)			3-fact (S_3)		
Layer	Within	Between	Ratio	Within	Between	Ratio
0	0.112	0.186	0.60	0.094	0.200	0.47
5	0.166	0.476	0.35	0.176	0.501	0.35
10	0.289	0.567	0.51	0.307	0.606	0.51
21	0.200	0.516	0.39	0.203	0.545	0.37
23	0.265	0.685	0.39	0.277	0.745	0.37

it (rotation of edges). The architecture is capable; the question is whether the data demands it.

(C) The reward. Next-token prediction does not incentivize algebraic coherence. Composing permutation operators is never required for correct prediction. A per-permutation linear map is sufficient and has strictly more capacity than a group-constrained representation ($n!$ independent matrices vs. matrices satisfying $O(n!^2)$ algebraic constraints).

The likely answer combines (A) and (C): the symmetry is not present in natural language, *and* the training objective would not reward learning it even in domains where it exists. Group structure in representations is not free—it requires either architectural inductive bias (equivariant layers), data with genuine symmetry (physical rotations), or an objective that rewards compositional generalization over group orbits.

5.2 Functional vs. Structural Equivariance

Our models achieve functional equivariance: swapping Alice and Bob reliably produces a swapped completion. But they do not achieve structural equivariance: the internal operators implementing this swap do not compose as a group. The model has learned a *lookup table*, not an *algebra*.

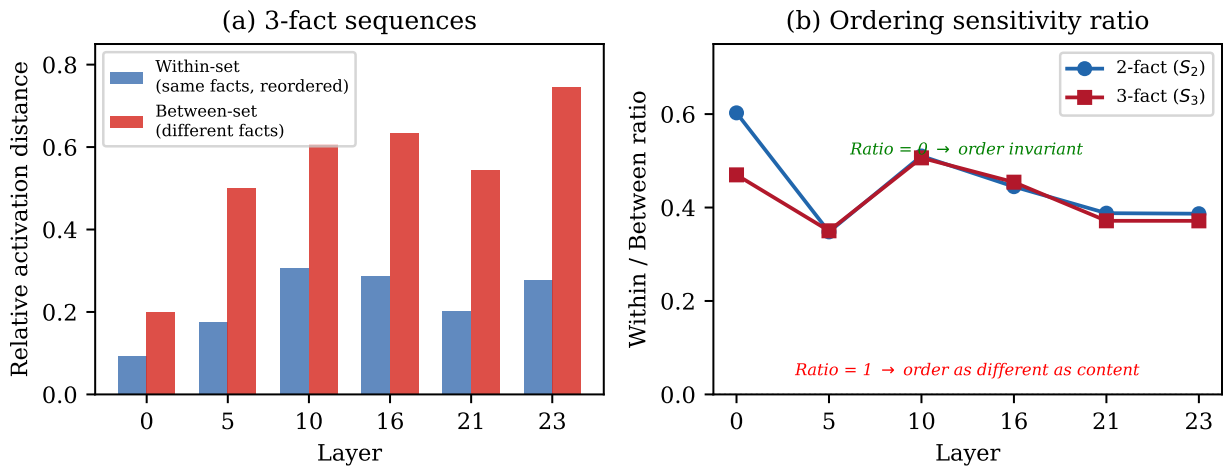

Figure 3: Impact of fact reordering on model representations. (a) Absolute activation distances for 3-fact sequences: reordering the same facts (blue) shifts representations 35–50% as much as entirely different facts (red). (b) The within/between ratio is stable at 0.35–0.50 across layers for both 2-fact and 3-fact sequences, far from 0 (order invariance).

Table 4: Behavioral impact of fact reordering. Top-1 agreement below 100% indicates the model's most likely next token changes when facts are reordered.

	Mean KL	Top-1 Agreement	Even KL / Odd KL
2-fact (S_2)	0.355	66.7%	—
3-fact (S_3)	0.285	70.7%	0.366 / 0.231

This distinction has a parallel in human cognition. Humans develop functional world models from temporally ordered experience, optimized by reward signals (dopamine, social feedback) that select for prediction accuracy rather than structural fidelity. The result is mostly-consistent representations with tolerance for cognitive dissonance—functional without being algebraically coherent. In physics, discovering that functional regularities arise from structural symmetries required stepping *outside* the local optimization landscape—a contribution of theoretical reasoning, not of the optimization process itself.

5.3 Implications for Synthetic Data Generation

Our results have three levels of implication for synthetic data design:

Level 1: Fact ordering is not a nuisance variable. Reordering the same facts changes model predictions 30% of the time. If synthetic training data presents facts in arbitrary order, it generates examples with different implicit salience signals. This is directly measurable and directly actionable.

Level 2: Product-space decomposition has blind spots. Frameworks like Simula [6] decompose data domains as $\mathcal{T}_0 \times \mathcal{T}_1 \times \cdots \times \mathcal{T}_K$ and sample factors independently. Our results show that trained models have learned structure that cannot be decomposed as a product of independent factors. Ordering, salience, and discourse structure create correlations between factors that product-space sampling misses.

Level 3: Symmetry is earned, not assumed. In physics, symmetries are *discoveries*—insights that required stepping outside local reward signals. In synthetic data generation, symmetries are *assumptions*—choices about what transformations preserve meaning. Our work suggests caution about assuming symmetries that have not been empirically verified. The Group Crosscoders result shows that group structure

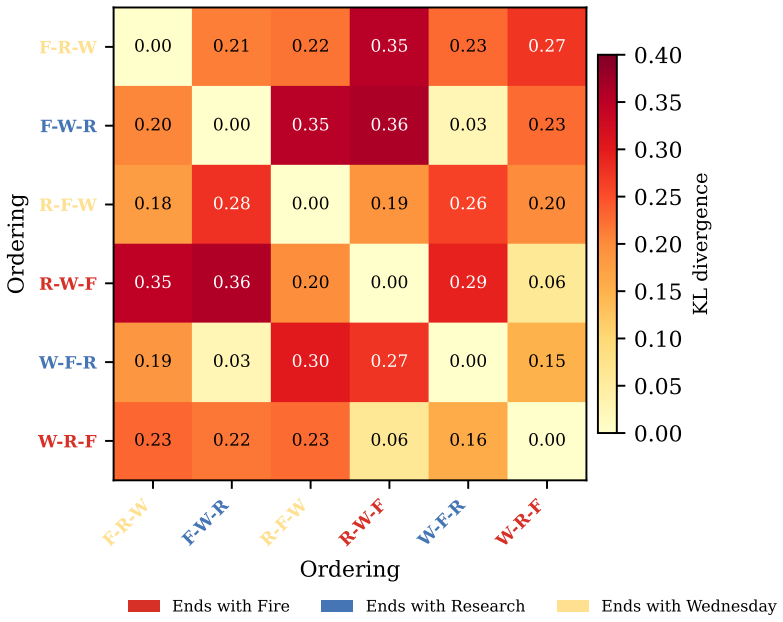

Figure 4: KL divergence between next-token distributions for all 6 orderings of three facts: "The room is on fire" (F), "I am researching LLM representations" (R), "It is Wednesday" (W). Orderings sharing the same final fact cluster together (block structure along anti-diagonal), revealing a recency effect. Tick labels are colored by final fact.

does emerge when the data has physical symmetry. Our result shows it does *not* emerge for linguistic permutations. The lesson: verify, don't assume.

6 Limitations

We tested only small symmetric groups (S_2, S_3); larger groups or continuous symmetries may behave differently. Our experiments use two models from the same family (Qwen2.5); architecture-specific effects are possible. The fact-level permutation experiments use relatively simple multi-sentence sequences; more complex discourse structures may show different patterns. We did not test whether training with permutation-augmented data would *induce* group structure, nor did we directly compare model performance when fine-tuned on ordered vs. shuffled synthetic data. Our positive control uses low-dimensional spaces where the system is well-conditioned; the analogy to high-dimensional neural activations is approximate.

7 Future Work

Several directions follow naturally: (1) Train models on permutation-augmented data and measure downstream impact on the presence of group structure and on task performance. (2) Test whether explicitly equivariant architectures develop group representations for linguistic permutations. (3) Extend to other symmetries in language: negation ($\mathbb{Z}_2$), tense shifts, semantic role reversals. (4) Investigate whether biological visual systems (via fMRI population codes) show rotational group structure, paralleling the Group Crosscoders finding in artificial vision models. (5) Examine weight matrices directly for emergent group structure, complementing our activation-level analysis.

8 Conclusion

Transformer language models develop permutation-sensitive but not permutation-equivariant representations. This is not a failure of the model—it reflects the structure of the task and the data. Idea space does not have S_n symmetry; fact ordering encodes salience, priority, and narrative structure. Local optimization under next-token prediction correctly learns this asymmetry without discovering the group-theoretic abstractions that would be needed for compositional generalization over permutations.

For synthetic data generation, this means that treating semantic content as order-invariant is an empirical assumption that may not hold. Ordering should be treated as a meaningful dimension of the data—a carrier of information about salience and discourse structure—rather than a nuisance variable to be averaged out.

More broadly, our results illustrate that the presence or absence of symmetry in neural representations is diagnostic of the presence or absence of symmetry in the data domain. Group Crosscoders find rotational structure in vision because rotational symmetry is real. We find no permutation structure in language because permutation symmetry is not. Symmetry in representations is earned by symmetry in the world—it is not a universal gift of neural network optimization.

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